

Tutorial 6

Properly features & prices

x_1	x_2	y
3	5	2
3	3	3

Questions

1. Write ridge regression Cost fn
2. Calculate Wt updates

Cost fn:

$$J = \|y - xw\|^2 + \lambda \|w\|^2$$

Slope

$$m_1 \rightarrow x_1$$

$$m_2 \rightarrow x_2$$

Linear eqns

$$2 = c + m_1 \cdot 3 + m_2 \cdot 5$$

$$3 = c + m_1 \cdot 3 + m_2 \cdot 3$$

$$\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} \begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}$$

$y \quad \quad x \quad \quad w$

$$w = (X^T X + \lambda I)^{-1} (X^T y)$$

$$X^T = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix}$$

$$X^T X + \lambda I = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{bmatrix} + \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+1 & 3+3 & 5+3 \\ 3+3 & 9+9 & 15+9 \\ 5+3 & 15+9 & 25+9 \end{bmatrix} + 1 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

final wts & bias are, $w_1 = 5$, $w_2 = 2$, $w_3 = 1$, $w_0 = 2$

$$\text{Margin} = \frac{2}{\|w\|}$$

$$X^T X + \lambda I = \begin{bmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 35 \end{bmatrix}$$

$$(X^T X + \lambda I)^{-1} = \begin{bmatrix} 89/95 & -18/95 & -8/95 \\ -18/95 & 41/95 & -24/95 \\ -8/95 & -24/95 & 21/95 \end{bmatrix}$$

$$X^T y = \begin{bmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$(X^T X + \lambda I)^{-1} (X^T y)$$

$$= \begin{bmatrix} 89/95 & -18/95 & -8/95 \\ -18/95 & 41/95 & -24/95 \\ -8/95 & -24/95 & 21/95 \end{bmatrix} \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$= \begin{bmatrix} 0.2421 \\ 0.7263 \\ -0.0105 \end{bmatrix} = \begin{bmatrix} c \\ m_1 \\ m_2 \end{bmatrix}$$

$$\hat{y} = 0.2421 + 0.72x_1 + -0.01x_2$$

$$\hat{y}_2 = 0.2421 + 0.72(3) - 0.01(5) = 2.35$$

$$\hat{y}_3 = 0.2421 + 0.72(3) - 0.01(2) = 2.37$$

Cost fn: $J = \|y - xw\|^2 + \lambda \|w\|^2$

$$x = \begin{bmatrix} 1 & 3.5 \\ 1 & 3.8 \end{bmatrix} \quad w = \begin{bmatrix} 0.2421 \\ 0.72 \\ -0.010 \end{bmatrix}$$

$$xw = \begin{bmatrix} 2.352 \\ 2.372 \end{bmatrix}$$

$$y - xw = \begin{bmatrix} -0.35 \\ 0.63 \end{bmatrix}$$

$$\|y - xw\|^2 = (-0.35)^2 + (0.63)^2 = 0.52$$

$$\|w\|^2 = c^2 + m_1^2 + m_2^2$$

$$= (0.2421)^2 + (0.72)^2 + (-0.010)^2$$

$$= 0.527$$

$$\|y - xw\|^2 + \lambda \|w\|^2 = 0.52 + 0.57$$

$$= 1.09$$